# Genuine Multipartite Entanglement Analysis

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Model                              Ground-state properties           GME summary
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Hubbard ring (1-D, PBC)            E0                : -4.60352630  Extracted n-spin RDM : n = 3
                                   E1                : -4.29999276
System parameters                                                   Purity:
-----------------                  <S2>              : -0.000000      gamma = Tr(rho^2)
Sites (L)          : 8             S                 : -0.0000
Electrons (Ne)     : 8             <Sz>              : 0.000000     Separable cut criterion:
Hopping (t)        : 1.0           Multiplicity 2S+1 : 1.0            gamma_A = 1  <->  rho = rho_A x rho_B
Interaction (U)    : 4.0           Spin state        : Singlet (S=0)
Mode               : uniform                                        C_GME (Troiani et al.):
n-body order       : 3             Double occupancy                   C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))
                                   ----------------
                                   <D> total         : 0.759406     Convex roof (mixed):
                                   <D> / site        : 0.094926       C_GME(ρ) = min Σ pᵢ C_GME(|ψᵢ⟩)


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# Summary: All 3-Body RDM Clusters

System: L=8, Ne=8, U=4.0, t=1.0 | $E_0$=-4.603526

| Sites | Purity | C_GME | C_LB | C_cayl | C_expo | C_UB | Status |
|---|---|---|---|---|---|---|---|

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Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map |  BEST
  C_UB = Upper bound (eigendecomp) | Status: GME  =has GME | NO GME=separable | MIXED=mixed state